Developer Salary Prediction

ML1 Task 2 — Regression

Łukasz Madej Osman Aliyev

May 2026

Problem

Goal: Predict annual developer compensation (`annual.pay.usd`)

Metric: Root Mean Square Error (RMSE)
Constraint: ML1-syllabus models only (no trees)

Training rows	~2 500
Test rows	628
Target	Annual salary in USD (heavy-tailed)
Key challenge	Multi-select survey columns, extreme outliers

Three decisions made before any model:

1. Multi-select columns (tech stack, tools, languages) — multi-hot + count features
2. Target modelled in **log-space** to stabilise variance
3. Salary outliers handled by **range filter** [1k, 300k]

Algorithms Considered

5-fold cross-validation on the same pipeline, RMSE in USD:

Model	Unfiltered data	After range filter
Ridge	93 363	39 482
Lasso	90 980	38 232
Elastic Net	91 848	38 702
KNN	87 898	38 826
SVR (RBF)	**86 295**	**36 659**
Random Forest *(reference)*	84 277	34 383
Gradient Boosting *(reference)*	84 194	34 658

- Range filtering alone cuts RMSE by ~55 k — extreme salaries dominate squared error
- **SVR is the best ML1-allowed model**; tree methods (out of scope) are only ~2 k better
- Linear models lag significantly — salary signal is non-linear

Feature Engineering

Multi-select columns (technologies, tools, languages, frameworks):

- Multi-hot encode each column (`min_freq = 100`, `top_k = 10` per column)
- Add **count feature** per column — breadth-of-experience proxy
- Derived: `professional_experience_share`, `non_professional_years`

Target encoding (8 high-cardinality columns):

- `region`, `industry`, `company.size`, `age.group`, `dev.role`, `education`, `employment.type`, `work.location`
- Smoothed K-fold target encoding — no leakage across folds

Preprocessing:

- StandardScaler + mean imputation
- Salary filter [1 000, 300 000] applied only to training data
- All steps fitted on training fold only

Hyperparameter Tuning

Three rounds of tuning for SVR:

1. **Preprocessing sweep** — scaler, imputer, `min_freq` / `top_k` for multi-hot encoding
2. **SVR grid search** — C in {1, 2, 3, 5}, epsilon in {0.03, 0.05, 0.08}, gamma in {scale, auto, 0.005, 0.01}
3. **Target encoding ablation** — progressively adding high-cardinality columns

Tuning stage	5-fold CV RMSE
Initial SVR (default params)	36 659
After preprocessing tuning	33 395
+ SVR grid search	31 488
+ Target encoding (8 cols)	**31 284**

Selected: `SVR(kernel=rbf, C=1, eps=0.03, gamma=auto)` with StandardScaler and mean imputation

Ensembling

Out-of-fold (OOF) blend of SVR + Random Forest + Ridge:

- Each model produces OOF predictions on the training set
- Weights optimised with Nelder-Mead on OOF RMSE
- Best found blend: **SVR 90.3% + RF 9.7%** (Ridge weight collapsed to ~0)

Configuration	OOF RMSE (eval on [1k, 300k])
SVR alone	31 284
SVR + RF + Ridge (Nelder-Mead)	**30 984**

Training on full data:

- Blend weights fixed from OOF optimisation
- Both SVR and RF retrained on all ~2 300 filtered rows
- Test predictions: 90.3% SVR + 9.7% RF (on log-scale, then exponentiated)

Why the Blend

Argument	Detail
OOF gain is real	-300 RMSE on 5 folds, consistent across splits
RF complements SVR	RF catches mid-range clusters SVR smooths over
Tiny RF weight	9.7% limits risk of overfitting RF to training noise
Ridge dropped	Ridge too weak (38k) to add value alongside SVR (31k)

Risks acknowledged:

- RF is outside the ML1 syllabus — used only as a blend component in the final submission
- OOF evaluation uses filtered salaries [1k, 300k]; true test set may include extreme values
- Expected ~2–3 k CV-to-LB gap: range filter before CV split makes CV optimistic

Bottom line: SVR alone at ~31 k is the "safe" submission; the blend at ~31 k is a ~300-point improvement with limited downside.

Expected Results

Final pipeline (end-to-end):

1. Salary filter [1k, 300k] on training data only
2. Multi-hot encode + count / derived features
3. K-fold target encode 8 high-cardinality columns
4. StandardScaler + mean imputation
5. **SVR** (RBF, C=1, eps=0.03, gamma=auto) fit on log-salary
6. Optional: blend 90.3% SVR + 9.7% RF predictions
7. `exp(prediction)` → submission

Estimate	Value
5-fold CV RMSE (filtered eval, SVR alone)	31 284
5-fold CV RMSE (OOF blend)	30 984
Expected leaderboard RMSE	~33–34 k
CV-to-LB gap	~2–3 k (structural, range filter artefact)

Summary

Best model	SVR with RBF kernel
Why SVR	Best ML1-allowed model; non-linear; robust to feature scale
Biggest win	Range filtering — removed extreme salaries (-55 k RMSE)
Second win	Smoothed K-fold target encoding (-5 k RMSE)
Blend gain	SVR 90.3% + RF 9.7% (-300 RMSE over SVR alone)
Expected LB	~33–34 k RMSE

Tried and rejected:

- Winsorization — hurt CV by ~4 k (clipped targets in validation folds)
- SVR + Ridge blend — Ridge too weak; blend pulled SVR down
- Stacking (meta-learner) — did not outperform the tuned single SVR

Near the data-driven ceiling for ML1-syllabus models on this dataset.